This Zenodo deposit contains a publicly available description of the Dataset:

"Untargeted lipidomic profiling of the liver in G2019S LRRK2 knockin mice under a specialized diet".

Dataset Description:

G2019S LRRK2 knockin mice and their wild-type littermates were maintained on a normal chow diet (23.7% kcal from fat) for the first 11 months of life. At that time, animals were randomly assigned to either a high-fat diet group (60% kcal from fat) or a control diet group (10% kcal from fat). Mice remained on their respective diets for an additional five months. At 16 months of age, mice were anesthetized with an isoflurane vaporizer, tissue were then collected and immediately snap frozen in liquid nitrogen. Frozen tissues were cryopulverized, homogenized and processed using the Bligh-Dyer method. Lipid identifications were assigned using LipidSearch (v5.0, Thermo Fisher Scientific) which was used to generate a compound list, and lipid peak areas were quantified in Skyline. Each experimental group contained biological replicates (n = 6 per group).

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Dataset Name: lee-mouse-ms-l-liver-g2019s-hf-diet, v0.1

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Publication DOI: NA

Grant IDs: ['ASAP-000592']

ASAP Lab: MOORE-LAB

ASAP Project: Are there bidirectional pathological interactions between senescence and LRRK2 models of PD

Project Description: Mutations in LRRK2 gene cause late-onset, autosomal dominant forms of familial PD that are clinically indistinguishable from sporadic PD. In addition, LRRK2 gene polymorphisms are associated with sporadic PD risk. The late disease-onset with these PD-linked genes suggest that interactions with age-related pathways are critical for manifesting or unmasking their pathogenic actions. Studies in experimental models further suggest that PD-linked LRRK2 mutations can induce abnormal mitochondrial function, lysosomal activity, autophagy, proteostasis and inflammatory processes, pathways that are dysregulated by cellular senescence. Our project focuses on determining whether:(1) PD-linked mutations in LRRK2 induce cellular senescence in the rodent brain, (2) increased senescence can exacerbate or unmask PD-like neurodegenerative phenotypes in LRRK2 mouse models, and (3) if removing senescent cells will provide neuroprotection in LRRK2 rodent models of PD.

Submission Date: 2026-04-30

This research was funded by the Aligning Science Across Parkinson's Collaborative Research Network (ASAP CRN), through the Michael J. Fox Foundation for Parkinson's Research (MJFF).